

Practical-3

Objective1: Write a program to implement the concept of inline function using two functions to convert the temperature from Celsius to Fahrenheit and vica versa.

Algorithm:

Step 1: Start.

Step 2: Define two inline functions:

celsiusToFahrenheit(c) to convert Celsius to Fahrenheit using:

$$F = 9/5 \times (C + 32)$$

fahrenheitToCelsius(f) to convert Fahrenheit to Celsius using:

$$C = 5/9 \times (F - 32)$$

Step 3: Input the temperature in Celsius.

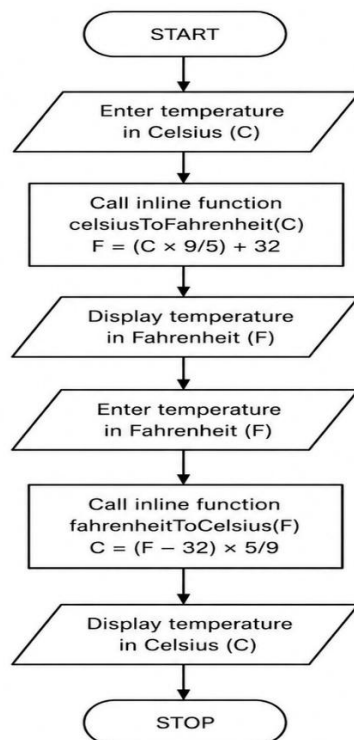
Step 4: Call the celsiusToFahrenheit() function and display the temperature in Fahrenheit.

Step 5: Input the temperature in Fahrenheit.

Step 6: Call the fahrenheitToCelsius() function and display the temperature in Celsius.

Step 7: Stop.

Flow Chart:



Input:

```
#include <iostream>

using namespace std;

inline float celsiusToFahrenheit(float c)
{
    return (c * 9 / 5) + 32;
}

inline float fahrenheitToCelsius(float f)
{
    return (f - 32) * 5 / 9;
}

int main()
{
    float c, f;

    cout << "Enter temperature in Celsius: ";

    cin >> c;

    cout << "Temperature in Fahrenheit: " << celsiusToFahrenheit(c) << " F" << endl;

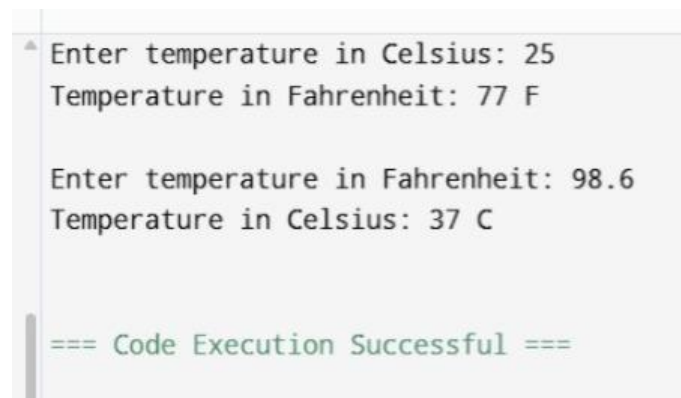
    cout << "\nEnter temperature in Fahrenheit: ";

    cin >> f;

    cout << "Temperature in Celsius: " << fahrenheitToCelsius(f) << " C" << endl;

    return 0;
}
```

Output:



```
Enter temperature in Celsius: 25
Temperature in Fahrenheit: 77 F

Enter temperature in Fahrenheit: 98.6
Temperature in Celsius: 37 C

=== Code Execution Successful ===
```

Objective 2: Write a program to implement the concept of default argument in a function for grading the students according to their five subject marks in which two subject marks are compulsory to be provided by the user.

Algorithm:

Step 1: Start.

Step 2: Define a function gradeStudent() with five parameters, where the last three parameters have default values of 70,80 and 50.

Step 3: Input the marks of the first two compulsory subjects.

Step 4: Call the function by passing only the first two subject marks.

Step 5: Inside the function:

Calculate the total marks.

Calculate the average marks.

If the average is greater than 90, display Grade A.

Else if the average is between 70 and 90 (inclusive of 70), display Grade B.

Otherwise, display Grade C.

Step 6: Input the marks of all five subjects.

Step 7: Call the function by passing all five subject marks.

Step 8: Again calculate the total, average, and display the appropriate grade according to the grading criteria.

Step 9: Stop.

Input:

```
#include <iostream>

using namespace std;

void gradeStudent(int m1, int m2, int m3 = 70, int m4 = 80, int m5 = 50){

    int total = m1 + m2 + m3 + m4 + m5;

    float average = total / 5.0;

    cout << "\nTotal Marks = " << total;

    cout << "\nAverage Marks = " << average;

    if (average > 90)

        cout << "\nGrade A";

    else if (average >= 70)
```

```

        cout << "\nGrade B";

else

        cout << "\nGrade C";}

int main(){

int m1, m2, m3, m4, m5;

    cout << "Enter marks of first two subjects: ";

    cin >> m1 >> m2;

    cout << "\nResult with only two subject marks:";

    gradeStudent(m1, m2);

    cout << "\n\nEnter marks of all five subjects: ";

    cin >> m1 >> m2 >> m3 >> m4 >> m5;

    cout << "\nResult with all five subject marks:";

    gradeStudent(m1, m2, m3, m4, m5);

return 0;}

```

Output:

```

Enter marks of first two subjects: 95
85

Result with only two subject marks:
Total Marks = 380
Average Marks = 76
Grade B

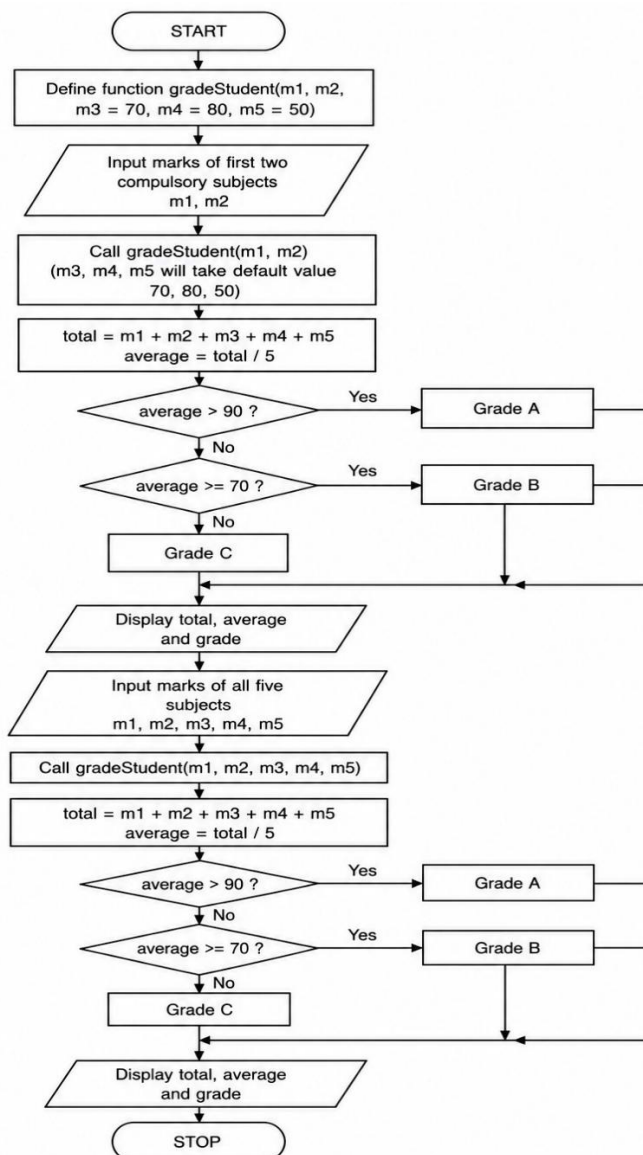
Enter marks of all five subjects: 60 85 90 89 95

Result with all five subject marks:
Total Marks = 419
Average Marks = 83.8
Grade B

=== Code Execution Successful ===

```

Flow Chart:



Objective 3: Wap to implement the concept of function overloading to check whether user is eligible for vote or not for a particular post as

1. check(int age) 2. check(int age, string name)

3. check(int age , string name, string post).

Algorithm:

Step 1: Start.

Step 2: Define three overloaded functions named check():

check(int age)

check(int age, string name)

check(int age, string name, string post)

Step 3: Input the age of the user.

Step 4: Call check(age).

If age $\geq$ 18, display "Eligible to Vote."

Otherwise, display "Not Eligible to Vote."

Step 5: Input the user's name and age.

Step 6: Call check(age, name).

Display the user's name.

If age $\geq$ 18, display "Eligible to Vote."

Otherwise, display "Not Eligible to Vote."

Step 7: Input the user's name, age, and the post for which voting eligibility is to be checked.

Step 8: Call check(age, name, post).

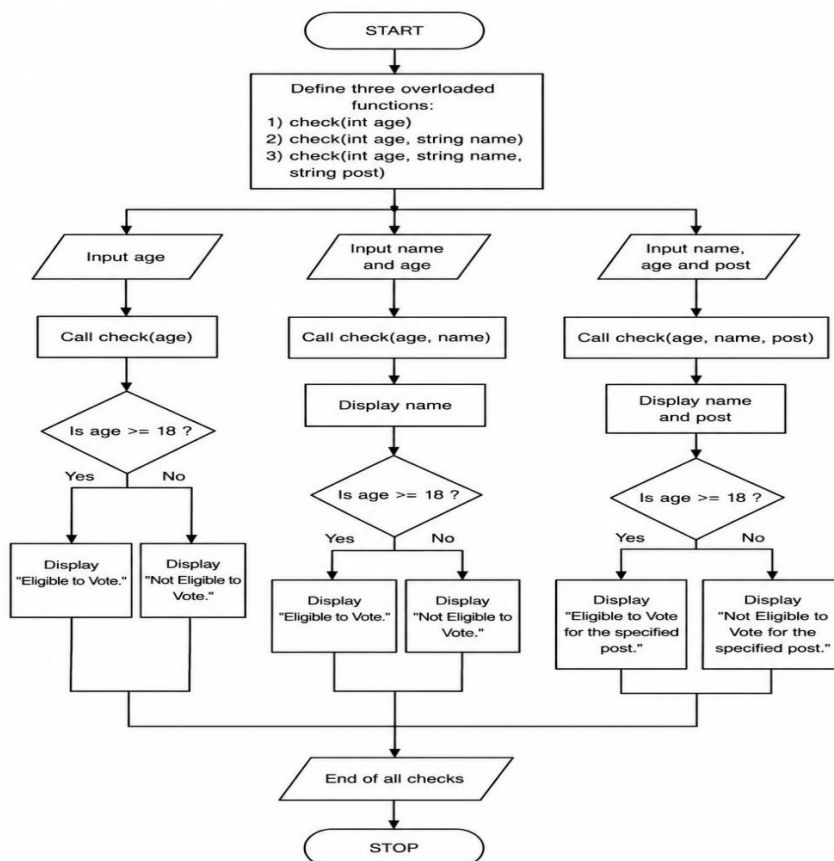
Display the user's name and post.

If age $\geq$ 18, display "Eligible to Vote for the specified post."

Otherwise, display "Not Eligible to Vote for the specified post."

Step 9: Stop.

Flow Chart:



Input:

```
#include <iostream>

#include <string>

using namespace std;

void check(int age){

    if (age >= 18)

        cout << "Eligible to Vote." << endl;

    else

        cout << "Not Eligible to Vote." << endl;}

void check(int age, string name){

    cout << "Name: " << name << endl;

    if (age >= 18)

        cout << "Eligible to Vote." << endl;

    else

        cout << "Not Eligible to Vote." << endl;}

void check(int age, string name, string post){

    cout << "Name: " << name << endl;

    cout << "Post: " << post << endl;

    if (age >= 18)

        cout << "Eligible to Vote for the post of " << post << "." << endl;

    else

        cout << "Not Eligible to Vote for the post of " << post << "." << endl;}

int main(){

    int age;

    string name, post;

    cout << "Enter Age: ";

    cin >> age;

    check(age);

    cout << "\nEnter Name: ";

    cin.ignore();

    getline(cin, name);
```

```
cout << "Enter Age: ";

cin >> age;

check(age, name);

cin.ignore();

cout << "\nEnter Name: ";

getline(cin, name);

cout << "Enter Age: ";

cin >> age;

cin.ignore();

cout << "Enter Post: ";

getline(cin, post);

check(age, name, post);

return 0;

}
```

Output:

```
Enter Age: 20
Eligible to Vote

Enter Name: Rahul
Enter Age: 24
Name: Rahul
Eligible to Vote

Enter Name: Shivank
Enter Age: 40
Enter Post: HR
Name: Shivank
Post: HR
Eligible to Vote for the post of HR.
|

=== Code Execution Successful ===
```